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To Keep or Not to Keep: Effects of Online Customer Reviews on Product Returns

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Abstract

While many studies examined the effects of online customer reviews (OCRs) on product sales, a clear understanding of the effects of OCRs on product returns is lacking. This study examines the impact of OCRs characteristics (valence, volume, and variance) on return decisions with a rich multi-year dataset from a major online retailer covering the electronics and furniture category. The main finding is that overly positive review valence (i.e., higher than the long-term product average), induces more purchases, but also more returns. An explanation for these findings is that OCRs help to form product expectations at the moment of purchase. Therefore, the purchase probability increases but the high expectations due to overly positive reviews may not be met, which results in negative expectation disconfirmation and consequently increases return probability as well. The effect of review valence on returns is stronger for novice buyers and for cheaper products. We further find that review volume and variance mainly affect purchase decisions, and have little to no effect on product returns. This study thus demonstrates that products returns should be considered when examining OCR effects, especially because overly positive reviews may hinder a retailer's financial performance, due to large reverse logistics costs associated with product returns.
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Keywords: Product returns; Online customer reviews; e-Commerce; e-WOM

Product returns are a severe, costly problem for retailers with substantial profit impacts. U.S. customers return a hefty $264 billion worth of products annually (Kerr 2013), such that a one percent decrease in the return rate could reduce annual reverse logistics costs by an average of $17 million for large retailers (Accenture 2011). In online retailing, return rates are an even more severe problem since a recent Wall Street Journal article indicates that around 30% of the purchases are returned (Banjo 2013), and reverse logistics costs range between $6 and $18 per product (The Economist 2013).

Most products are not returned because of product defects but customers' negative post-purchase product evaluation (The Supply Chain Consortium 2008). For example in the electronics category, only five percent of the returns are related to defective products (Accenture 2011). Negative post-purchase product evaluations often arise due to the customers' limited ability to evaluate and test products before purchasing them. This limited ability creates uncertainty about product performance prior to purchase (Weathers, Sharma, and Wood 2007), which then increases the likelihood that a product will fail to meet customers' expectations (Rust et al. 1999). These unmet customer expectations result in dissatisfaction with the product and a higher return likelihood.

Customers' ability to adequately evaluate products before purchase is affected by the information provided by the retailer which thus affects the return rate (Bechwati and Siegal 2005). The costly problem of product returns has led retailers to invest in technologies such as zoom features to help customers to make better decisions and to avoid returns (De, Hu, and Rahman 2013). An additional source of information that is available on retailers' websites is online customer reviews (OCRs). OCRs complement retailer-provided information (Chen and Xie 2008), and may

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help to form customer expectations prior to purchase, and thus may affect return rates.

A large set of studies have demonstrated the strong impact of OCRs, and other types of electronic Word-of-Mouth, on sales such as confirmed by recent meta-analyses (Babic et al. 2016; Floyd et al. 2014). However, despite the importance of product returns and considerable interest in OCRs, academics and practitioners suffer from limited knowledge about the effects of OCRs on customers' product return decisions. Such knowledge is critical, when considering that predictions about the effects of OCRs on retailer performance may be overly optimistic or pessimistic if *only* the effects on sales are taken into account. It is thus the aim of this study to examine the net effect of OCRs, taking into account both the effects on customers' purchase and return decisions. We use product page views and transaction data that we gathered from a major European online retailer between 2011 and 2013, which include 8,835,645 page views that resulted in 631,063 purchase transactions for 2,164 different products in electronics and furniture categories. This unique and rich dataset allows us to examine the effects of the key characteristics (i.e., valence, volume, and variance) of OCRs available at the moment of purchase on customer purchase and return decisions by means of the page view and transaction data. Because only customers who purchased can make a return decision, we model the purchase and return decision jointly with a bivariate probit model that ensures that the return decision is made conditionally on the purchase decision and controls for product and customer heterogeneity.

A key challenge in attributing the effect of OCRs on the purchase and return decision is endogeneity. Products with better perceived quality may have more favorable OCRs, are more likely to be purchased and are less likely to be returned. Since we have longitudinal transaction data, we observe purchases and returns for the same product over a longer time period while the OCRs available for that product vary as new reviews are being posted. As such, we test how within-product variations in OCRs available over time affect the product purchase and return decision. We, furthermore, account for perceived product quality reflected in between-product average OCR variables. Moreover, we control for product and customer characteristics to account for common shocks that may influence OCRs, purchase and return probability. Based on our model and additional financial data provided by the online retailer, we simulate both the net effect (i.e., purchases–returns) and also the financial impact of OCRs on a retailer's gross margin.

The key insight of our study is that the OCRs available at the moment of purchase affect both the purchase and return decision. More specifically, if reviews are *overly positive* (i.e., current OCR valence is higher than the long term product average), this leads to more purchases but also increases the return probability. As such, review valence effects go beyond the moment of purchase, and these effects of review valence are substantial, a one point increase in review valence results in an increased purchase probability of 9.14% (electronics) and 14.60% (furniture) and increased return probability by 11.16% (electronics) and 10.34% (furniture). More notably, the effect of overly positive reviews on retailer's financial performance is negative, because of the high reverse logistics costs associated with product returns. We further find that the effect of review valence on returns is contingent on several factors. First, the effect of review valence on the return decision is weaker for customers experienced in buying at this retailer and in the specific category. Second, the effect of review valence on the return decision is also weaker for more expensive products. Regarding the effects of the other OCR characteristics, review volume increases the purchase probability but does not affect the return probability. Variance in review ratings significantly reduces the purchase probability and increases the return probability, though this latter effect occurs in only one of the two product categories. Finally, we show that OCRs reflect product quality; products with higher average long-term ratings (between product–effect) have a higher purchase and lower return probability. Thus, we establish that it is essential to account for the effects of OCRs on customers' return decisions.

With this study, we contribute to the literature by demonstrating that OCR effects go beyond the moment of purchase and also affect decisions to return or keep purchased products. There is only one study that also examines the effects of OCRs on product returns (Sahoo, Dellarocas, and Srinivasan 2016). Our study differentiates in two important aspects from this working paper. First, we consider valence, volume, and variance of OCR explicitly, while Sahoo, Dellarocas, and Srinivasan (2016) only consider volume and variance of OCR and use valence as a control variable. However, recent meta-analyses (Babic et al. 2016) stressed the relevance of OCR valence and, thus, it is critical to explicitly consider OCR valence. Second, we include both the within- and between-product variations of OCR in our model. We show that the positive effect of review valence on the return decision (i.e., within-product effect) contrasts the product quality effect reflected in review ratings (i.e., between-product effect): products with higher average long-term ratings have a higher purchase and lower return probability. The results that can be compared between both studies are consistent, but our approach provides richer insights by addressing all three OCR characteristics and by including both within- and between-product effects.

Theoretical Background

Customers' decisions to purchase and return a product are based on their level of expectations about the product's performance and the uncertainty surrounding these expectations. Uncertainty arises because customers do have incomplete information about the product when purchasing online (Wood 2001), so their expectations are imperfectly informed (Golder et al. 2012). Both customers' level of expectations about product performance and the attached uncertainty together determine customers' expected product utility (Rust et al. 1999). Customers decide to purchase a product if the expected utility is greater than the utility of not buying it (McFadden 1974). The level of expectations affects customers' purchase probability positively, whereas uncertainty reduces the purchase probability because customers are generally risk averse (Rust et al. 1999).

However, the information provided at the moment of purchase does not resolve the product at full and so customer

base their purchase decision on imperfect information (Shulman, Cunha, and Saint Clair 2015). After the purchase, the customer inspects the product and ultimately the product is revealed at full. If the product does not meet the expectations formed at the moment of purchase, the customer is dissatisfied and hence is more likely to return the product (Bechwati and Siegal 2005). That is, post-purchase satisfaction is a result of customer expectations formed at the moment of purchase, product performance, and the difference between them, similar to the expectation disconfirmation mechanism used to explain customer satisfaction (see Oliver 2009). A negative expectation disconfirmation (i.e., product performance lower than expected) lowers satisfaction and increases customer's return probability and vice versa.

As such, customers' level of expectations affects satisfaction in two ways and thereby also product return decisions. On the one hand, expectations have a direct positive effect on satisfaction, because customers are reluctant to acknowledge discrepancies from their existing beliefs (Oliver 2009). On the other hand, expectations have a negative indirect effect on satisfaction because the likelihood of negative expectation disconfirmation increases with expectations and thus lowers customer satisfaction (Szymanski and Henard 2001). These contrasting effects produce an ambiguous net effect of customer expectations on satisfaction, though a meta-analysis has suggested that the indirect negative effect outweighs the direct positive effect (Szymanski and Henard 2001). That is, the higher customers' expectations, the higher the product return probability when product performance remains constant.

Customers' expectations are characterized by uncertainty because these expectations are usually imperfectly informed at the moment of purchase in online settings (Wood 2001). Greater uncertainty in expectations may decrease satisfaction and consequently increase return probability because the effect of failing or exceeding expectations is asymmetric. The effect of failing expectations is stronger than the effect of exceeding expectations (i.e., losses loom larger than gains (Rust et al. 1999)) and hence, more uncertainty increases return probability. Taken together, a higher level of expectations should lead to higher purchase and return probabilities; more uncertainty in expectations should produce lower purchase and higher return probabilities, controlling for product and customer characteristics.

Effects of OCRs on Purchase Decision

Although customers form expectations about a product by using multiple information such as previous product experiences (Zeithaml, Berry, and Parasuraman 1993), price and brand name (Kirmani and Rao 2000), we focus on how OCRs available at the moment of purchase determine both the level and uncertainty of expectations. Customers form their pre-purchase product expectations partly on OCRs, and OCRs, thus, influence the purchase decision. Generally, OCRs are characterized by their valence, volume, and variance (e.g., Kostyra et al. 2016; Moe and Schweidel 2012). A large body of studies examined the effects of OCRs on the purchase decision. Recent meta-analyses indicate that review valence and volume have a positive effect on customer purchases (Babic et al. 2016; Floyd et al. 2014). Reviewers' overall product evaluations (i.e., valence) will affect the customer expectations related to the product (Li and Hitt 2008) whereas more reviews (i.e., volume) reduce uncertainty about expectations (Ho-Dac, Carson, and Moore 2013). Reviewers' dissensus (i.e., variance) has a negative effect on customer purchase decision (Babic et al. 2016; Floyd et al. 2014) because it decreases expectations and increases the associated uncertainty due to a broader range of positive and negative product evaluations (Chen and Lurie 2013; Khare, Labrecque, and Asare 2011). Given the focus of this study and the number of studies regarding the effects of OCRs on the purchase decision, we only develop hypotheses on the relationships between OCR characteristics and product return probability.

Effects of OCRs on Return Decision

If a customer purchases the product, the customer may decide to return the product if the expectations formed at the moment of purchase are not met. Since a customer's expectations are partly formed by the OCRs, OCRs available *at the moment* of purchase may affect the return decision. Review valence (i.e., average product rating) represents other customers' overall product evaluations (Li and Hitt 2010) and reflects their subjective evaluations of product performance. A customer searching for product information may partly form his or her own expectation level at the moment of purchase, in accordance with review valence (De Langhe, Fernbach, and Lichtenstein 2016; Kuksov and Xie 2010). Thus, customer's preference for a product may be formed or altered from the exposure to positive (negative) valence reviews, as valence is indicative for the expected product quality (Babic et al. 2016). When reviewers' evaluations increase for a product, customer expectations related to the product may rise compared to the same product having lower valence reviews (Li and Hitt 2008). Consequently, once the product has been purchased and inspected at home it may not meet these high expectations formed at the moment of purchase due to the reviews. Hence, negative expectation disconfirmation and dissatisfaction is more likely with higher review valence compared to lower review valence for that product (Szymanski and Henard 2001), which should increase return probability. Accordingly, we hypothesize that higher valence reviews increases return probability.

H1. Review valence has a positive effect on return probability.

Review volume (i.e., number of reviews) affects the uncertainty of customer expectations, in that it reflects the amount of information available (Archak, Ghose, and Ipeirotis 2011; Khare, Labrecque, and Asare 2011). Therefore, the number of reviews reflects the diagnosticity of OCRs (Floyd et al. 2014). More reviews at the time of purchase imply that there is more information available about the product which reduces uncertainty about expectations compared to that product having fewer reviews (Ho-Dac, Carson, and Moore 2013; Rust et al. 1999). Furthermore, the number of reviews delivers information about how many other people have experienced the product, which reduces uncertainty about the product further (Babic et al. 2016). Lower uncertainty then decreases the likelihood of negative

expectation disconfirmation after purchase, leading to higher levels of post-purchase satisfaction (Rust et al. 1999) and lower return probabilities. Accordingly, review volume is expected to lower return probability.

H2. Review volume has a negative effect on return probability.

Review variance (i.e., the variance in ratings) represents the degree of consensus across reviews of a product. With greater review variance at the moment of purchase, a customer reads a broader range of positive and negative product evaluations. The negativity bias suggests that negative reviews have a stronger impact on customers' expectations than positive reviews (Chen and Lurie 2013). High variance implies that a substantial number of product reviews is negative, and, thus, customer expectations may be lowered. Therefore, some customers may exclude the product from consideration and may not purchase it due to this high variance (Wang, Liu, and Fang 2015). Those customers, who purchase the product despite the high variance in ratings, may be uncertain about their expectations because of the incongruent information available about the product (He and Bond 2015; Khare, Labrecque, and Asare 2011). The increased uncertainty caused by review variance, moreover, increases the likelihood of negative expectation disconfirmation after the purchase which lowers post-purchase satisfaction (Rust et al. 1999), and increases return probability. Therefore, we postulate:

H3. Review variance has a positive effect on return probability.

Impact of Customer Experience and Product Price on the Effectiveness of OCR

When customers purchase online, their level and uncertainty in the expectations about the product play a role in the decision making (Shulman, Cunha, and Saint Clair 2015). The extent to which expectation disconfirmation plays a role may vary across customers and products. We focus on the moderating effects with review valence because review valence affects customers' level of expectations (Kuksov and Xie 2010). Therefore, we examine whether customer experience and product price moderate the effect of review valence on the return decision.

Customers form their expectations based on multiple information sources, the most prominent source is previous experiences (Zeithaml, Berry, and Parasuraman 1993). If such internal information is not available, external information such as OCRs become especially relevant to inform or alter the expectations (Babic et al. 2016). Customers, who have recently bought from the focal retailer in a specific category, may instead rely less on review valence. These customers count more on their own experiences to form product expectations (Rust et al. 1999). In addition, the retailer-provided information focuses on technical specifications which are more informative for experienced customers (Chen and Xie 2008). Hence, review valence may provide less additional information in helping to identify if the product matches the customer needs for experienced customers. As a result, OCR ratings are less informative for experienced customers to form their expectations (Chen and Xie 2008). Experienced customers are those customers who have recently bought from the focal retailer in a specific category. Therefore, to be in line with expectation-disconfirmation mechanism, the effect of review valence should be weaker for experienced customers.

H4. The positive effect of review valence on the return decision is weaker for experienced customers (i.e., customers who have recently bought from the focal retailer in the focal category).

The effect of review valence on the return decision may also be moderated by product price. For more expensive products, product involvement is higher because there is a higher risk associated with purchase (Prelec and Loewenstein 1998). Because customers have a greater concern about making the 'best' purchase decision, the effect of valence on the purchase decision is stronger for higher priced products (Floyd et al. 2014). In addition, previous studies found higher return rates for higher priced products because customers are more concerned about making a good decision when receiving the product (Petersen and Kumar 2009). Therefore, for higher priced products customers spend a greater time on making the decision after the purchase and rely less on the expectations formed at the moment of purchase (Petersen and Kumar 2009). Accordingly, we expect that the effect of review valence on the return decision is weaker for higher priced products.

H5. The positive effect of review valence on the return decision is weaker for higher priced products.

Data

The data for this study came from a major European online retailer[4] that sells exclusively through its online store and the provided data encompassed two product categories: electronics and furniture. We gathered daily product page views, product purchases, and returns, as well as product-specific OCR information from January 2011 to June 2013. Because we are interested in the effects of OCRs on customers' return decisions, we only consider products that invoked OCRs during the study period (Chevalier and Mayzlin 2006). Moreover, we include only those products that prompted at least 30 purchases during the observation period, to ensure sufficient within-product variation in the return probability.[5] In this study, a product may encompass several SKU's, for example a specific type of mobile phone available in different colors is categorized as one product in line with how products and associated reviews are displayed at the retailer's website. For the set of products, we included the page views and purchases in the database. The final database contains information about 8,835,645 page views (electronics 6,696,578; furniture 2,139,067) and 631,063 purchases (electronics 552,619; furniture 78,444), related to 2,164 products (electronics 1,722; furniture 442). We use information about each customer's product page views to model the purchase decision; the product page view data came from server logs that captured product page requests. The average conversion rate is

[4] The retailer requested to remain anonymous.

[5] We also estimated models with a minimum of 40 transactions. The general findings remained the same, and are available on request.

Table 1
Descriptive statistics.

	Electronics	Furniture
Panel A: Product characteristics		
Product purchase rate (%)		
Mean	9.41	4.05
Std. dev.	8.67	3.55
Product return rate (%)		
Mean	8.35	10.06
Std. dev.	4.90	5.01
Years product is on market		
Mean	.80	1.81
Std. dev.	1.03	1.43
Product price (in EUR)		
Mean	276.27	287.07
Std. dev.	272.33	231.55
Number of products	1,722.00	442.00
Panel B: OCR characteristics		
Review valence		
Mean	4.48	4.25
Within-product std. dev.	.16	.20
Between-product std. dev.	.53	.64
Range	1.00–5.00	1.00–5.00
Review volume		
Mean	9.46	6.50
Within-product std. dev.	9.58	4.04
Between-product std. dev.	6.85	5.40
Range	0.00–167.00	0.00–69.00
Panel C: Review rating distribution		
Five-star reviews	53.9%	44.4%
Four-star reviews	39.7%	44.9%
Three-star reviews	4.4%	7.1%
Two-star reviews	1.3%	2.5%
One-star reviews	0.8%	1.1%

9.41% in electronics and 4.05% in the furniture category. The average time a product remained available in the online store differed between categories, from .80 years for electronics to 1.81 years for furniture (Table 1, panel A). The continual introduction of new products increased the uncertainty of customer expectations prior to purchase (Anderson, Karsten, and Simester 2009).

To model customers' return decisions, we used information about whether a purchase was returned or not (Anderson, Karsten, and Simester 2009; Petersen and Kumar 2009). The average return rates (electronics = 8.35%, furniture = 10.06%; Table 1, panel A) correspond with overall industry estimates (e.g., nine percent, Kerr 2013). Customers could return a purchase for up to two weeks after they received the product delivery, without needing to offer reasons and at no cost. This policy includes heavy and bulky items that the retailer picks up from the customer's address. The online retailer's return policy did not change during our study period.

Customer reviews feature an overall product evaluation, using a discrete five-star rating, as is common online (Moe and Schweidel 2012). The average review rating and number of reviews are prominent, appearing directly below the product name. Individual star ratings and the related text appear, by default, in reverse chronological order, below the product description, which also is common in online settings (Moe and Schweidel 2012). The retailer publishes newly posted reviews on its website once per day, because before publishing the reviews, reviews are checked whether the content is not illegal, invasive for privacy, or contains any form of unsolicited commercial message.

Table 1 (panels B and C) presents the descriptive statistics for OCRs per product category. On average, the volume of reviews per product was greater in the electronics than in the furniture category (electronics $M = 9.46$, furniture $M = 6.50$). The ratings posted at this retailer are typical of most posted reviews online in that the reviews are overwhelmingly positive (average star rating: electronics $M = 4.48$, furniture $M = 4.25$), more than 80% earn four- or five-star ratings (see also Chevalier and Mayzlin 2006; Moe and Trusov 2011).

To test our hypotheses of OCRs on the return probability, we use the within-product variation over time because these variables do not contain the perceived product quality and popularity effects. Hence, we decomposed the variation in star ratings into within-product variation (electronics SD = .16, furniture SD = .20) and between-product variation (electronics SD = .53, furniture SD = .64). That is, we found both variation in review ratings within a product over time and heterogeneity in the average star ratings between products. The between-product variation in OCRs are good proxy characteristics of the product, where the average rating reflects perceived product quality, the average volume reflects popularity and the average variance reflects heterogeneity in fit with the customer base.

In H4, we test whether customer experience moderates the effect of OCRs on the return decision. At this retailer, customers make relatively few repeat purchases in the electronics and furniture categories; in electronics, 33.3% of the transactions and in furniture 13.2% of the transactions are made by customers who purchased in the category during the last year. This low purchase frequency suggests substantial uncertainty prior to purchase because few customers purchased recently at the retailer in these categories (Petersen and Kumar 2009). In H5, we test whether the effect of valence is moderated by product price. The average product price is relatively similar across the two product categories (electronics $M = €276.27$, furniture $M = €287.07$, see Table 1 panel A).

Model Formulation

Purchase and Return Decisions

Given the research objective of this study, we need a model that examines the return decision. However, only customers who purchase can make the return decision and hence the model should take into account that the return decision is conditional on the purchase decision. In this model, customer i decides to purchase product j at day t if the utility of purchasing ($U_{ijt}^{purchase}$) is higher than the utility of not purchasing (McFadden 1974; Rust et al. 1999). After making a purchase, a customer returns the product if the utility of returning the product (U_{ijt}^{return}) is larger than the utility of keeping the product. The utility of returning

a product depends on customers' expectation (dis)confirmation and thus (dis)satisfaction with the product.

The utilities of purchasing and returning the product can both be divided into observed ($V_{ijt}^{purchase}$, V_{ijt}^{return}) and unobserved ($\varepsilon_{ijt}^{purchase}$, $\varepsilon_{ijt}^{return}$) components. Assuming that customers maximize their utility when making the purchase/return decisions and that the unobserved components of the utilities follow a bivariate normal distribution, we employ a bivariate probit formulation with endogenous sample selection (Greene 2012, pp. 789–791). The bivariate probit model with endogenous sample selection ensures that the return decision is not made independently but *conditionally* on the purchase decision. In this model, there are three possible outcome probabilities, namely no purchase (1), purchase and return (2) and purchase and keep (3):

$$Pr(Purchase_{ijt} = 0 \,|\, X_{ijt}^{purchase}) = \Phi(-V_{ijt}^{purchase}) \tag{1}$$

$$Pr(Return_{ijt} = 1 \,|\, Purchase_{ijt} = 1, X_{ijt}^{return}, X_{ijt}^{purchase}) \cdot Pr(Purchase_{ijt} = 1 \,|\, X_{ijt}^{purchase}) = \Phi_2(V_{ijt}^{return}, V_{ijt}^{purchase}, \rho) \tag{2}$$

$$Pr(Return_{ijt} = 0 \,|\, Purchase_{ijt} = 1, -X_{ijt}^{return}, X_{ijt}^{purchase}) \cdot Pr(Purchase_{ijt} = 1 \,|\, X_{ijt}^{purchase}) = \Phi_2(-V_{ijt}^{return}, V_{ijt}^{purchase}, -\rho) \tag{3}$$

$$\begin{pmatrix} \varepsilon_{ijt}^{purchase} \\ \varepsilon_{ijt}^{return} \end{pmatrix} \,\Big|\, X_{ijt}^{purchase}, X_{ijt}^{return} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right] \tag{4}$$

where Φ is the univariate standard normal c.d.f., Φ_2 is the bivariate standard normal c.d.f., ρ is the correlation between the error terms of the purchase and return decisions (Eq. (4)) and $X_{ijt}^{purchase}$, X_{ijt}^{return} are the observed covariates that affect the purchase and return decisions, respectively.

Model Specification: OCR

We argue that OCRs available at the moment of purchase affect the utilities of purchasing ($V_{ijt}^{purchase}$) and returning (V_{ijt}^{return}) the product. We consider the following OCR characteristics for product j at time t; the average rating ($valence_{jt}$), the number of reviews ($volume_{jt}$) and the variance in product ratings ($variance_{jt}$). The OCR variables focal in our study are within-product variables of reviews (i.e., Mundlak correction; Greene 2012), which are operationalized as the current value minus the product average (e.g., $valence_{jt} - \overline{valence_j}$). We thus assess the within-product effects of OCR (e.g., $valence_{jt} - \overline{valence_j}$) which are focal for hypothesis testing and include the between-product effects of reviews (e.g., $\overline{valence_j}$) to account for perceived product quality. Similarly, we account for product-level average volume and variance, to control for between-product variation in popularity and heterogeneity.

For the within-product effects of reviews (e.g., $valence_{jt} - \overline{valence_j}$), we consider all reviews published prior to the day of the focal purchase to ensure that a customer could actually observe it to determine the effects of OCRs on customers' purchase and return decision. Initially, a product has no reviews. Thus, we imputed values of zero for the within-product effects of valence and variance. All imputations equal the product mean, because we used product–mean centering.

We only consider the within-product effects of OCRs because we believe that there can be two sorts of endogeneity in OCRs. First, products that induce on average higher valence offer greater appeal, which may explain why they are purchased more often (Moe and Trusov 2011). To distinguish this perceived product quality effect from the effect of OCRs itself, we include the overall product average of the OCR characteristics for our observation period as additional variables in the utilities for purchasing and returning (e.g., $\overline{valence_j}$; Mundlak 1978) to account for this possible source of endogeneity. We do so, because the average rating over the product life time and the number of reviews correlate with product quality (De Langhe, Fernbach, and Lichtenstein 2016). Thus, in this way we separate out the potential endogeneity of perceived product quality reflected in OCRs.

A second source of endogeneity is that product quality perceptions varying for a product over time may influence both the availability of OCRs and the return probability for a product. To account for this source of endogeneity, we considered other *product characteristics* which may affect customers' decision to return the product. The quality perceptions may vary with the product price ($price_{jt}$) and the time the product is on the market (YOM_{jt}). Furthermore, we included the ratio of products introduced in the product category since the introduction of the focal product ($new_competitors_{jt}$) because newly introduced products may affect the quality perceptions for the focal product. In addition, we considered the average unit sales ($\overline{sales_j}$) as a product characteristic to capture a product's popularity. We include the question ($\overline{questions_{jt}}$) and the fraction of questions answered ($frac_answered_{jt}$) about the product as controls because if a product receives many questions this may indicate that there is a problem with the product or with the product description. These Q&A sections are displayed on the product page; the questions appear below the reviews, the answer appears directly below the question which is the formatting used by most online retailers. In these sections, customers can raise questions that other customers or the retailer might answer. We again include both the within effect (e.g., $\overline{questions_{jt}} - \overline{questions_j}$) and the between effect as controls (e.g., $\overline{questions_j}$). Thus, in this way, we separate out the potential time-varying perceived product quality reflected in variations of OCRs within the product.

Model Specification: Control Variables and Utility Function

Next to the *product characteristics* discussed above that we include as controls, we included *customer characteristics* that may also affect a customer's purchase and return decision. Specifically, we considered a customer's age (age_{it}), gender ($male_i$), area of living ($urbanicity_{it}$), length of relationship with

Table 2
Variable operationalization in the purchase and return decision model.

Variable	Description and measure
$purchase_{ijt}$	Variable indicating whether product j on day t is purchased by customer i
$return_{ijt}$	Variable indicating whether product j on day t is returned by customer i
Within-product effects of OCRs	
$valence_{jt} - \overline{valence_j}$	Average of all ratings for product j posted before day t minus average valence of product ratings for product j
$volume_{jt} - \overline{volume_j}$	Number of reviews for product j posted before day t minus average number of reviews for product j
$variance_{jt} - \overline{variance_j}$	Variance across all ratings for product j posted before day t minus average variance of product ratings for product j
Between-product effects of OCRs	
$\overline{valence_j}$	Average review rating for product j
$\overline{volume_j}$	Average number of reviews for product j
$\overline{variance_j}$	Average variance in review rating for product j
Product characteristics	
$price_{jt} - \overline{price_j}$	Price of product j on day t (in Euros) minus average price of product j
$YOM_{jt} - \overline{YOM_j}$	Time product j is on the market at day t (in years) minus average time product j is on the market
$\overline{price_j}$	Average price for product j (in Euros)
$\overline{YOM_j}$	Average number of years product j is on the market (in years)
$questions_{jt} - \overline{questions_j}$	Number of questions for product j posted before day t minus average number of questions about product j
$frac_answered_{jt} - \overline{frac_answered_j}$	Fraction of answered questions for product j before day t minus average fraction of answered questions for product j
$new_competitors_{jt}$	Number of products introduced in the product category on day t since the introduction of product j over all products in category
$\overline{sales_j}$	Average sales for product j (in units/week)
$\overline{questions_j}$	Average number of questions for product j
$\overline{frac_answered_j}$	Average fraction of questions answered for product j
Customer characteristics	
age_{it}	Age of customer i (in years) on day t
$male_i$	Customer i is male (1) or female (0)
$urbanicity_{it}$	Customer i lives in an urban area, 1 (urban)–5 (rural) on day t
$relationship_length_{it}$	Time since first purchase made by customer i (in years) on day t
$previous_purchase_{it}$	Customer i made a purchase (1), or did not make a purchase (0) in the category of product j during the last 12 months on day t
$previous_return_{it}$	Customer i made a return (1), or did not make a return (0) in the category of product j during the last 12 months on day t
$order_size_{it}$	Number of products bought by customer i on day t
Situation-specific variables	
$campaign_t$	Retailer runs a major marketing campaign on day t
$holiday_t$	Day t is in November or December
$category_pageviews_{jt}$	Total number of page views in category of product j on day t

the retailer ($relationship_length_{it}$), and indicators if the customer purchased during the last twelve months ($previous_purchase_{it}$), and returned during the last twelve months at the retailer ($previous_return_{it}$) (Petersen and Kumar 2009). Since Griffis et al. (2012) report larger order sizes by customers who return products, we also included number of items purchased in this specific order ($order_size_{it}$) in the return decision model.

Finally, *situation-specific* factors may attract different customers to buy or may affect the return decision and are therefore included as controls: major marketing campaigns that are not product-specific ($campaign_t$) and seasonal patterns ($holiday_t$). Moreover, we used the total number of page views for the category on day t to reflect general interest in the category across customers ($category_pageviews_{jt}$) only in the purchase decision. We do so because category popularity may affect the likelihood of buying a product in that category but seems theoretically unrelated to the return decision for one specific product. Thus, the number of category page views served as an exclusion criterion to reduce the correlation between the two error terms of the bivariate probit model (Greene 2012).

Thus, the utility that customer i purchases product j at day t and the utility that customer i returns product j bought at day t can be expressed as:

$$U_{ijt}^{purchase} = \beta_1 OCR_{jt} + \beta_2 product_{jt} + \beta_3 customer_{it} + \beta_4 situation_t + \beta_5 category_pageviews_{jt} + u_{1j} + \varepsilon_{ijt}^{purchase} \quad (5)$$

$$U_{ijt}^{return} = \delta_1 OCR_{jt} + \delta_2 product_{jt} + \delta_3 customer_{it} + \delta_4 situation_t + u_{2j} + \varepsilon_{ijt}^{return} \quad (6)$$

where OCR_{jt} denotes the vector of OCR variables, $product_{jt}$ is the vector of product related variables, $customer_{it}$ is the vector of customer related variables and $situation_t$ is a vector of situation related variables, all these variables are detailed in Table 2. β_{1-5} and δ_{1-4} denote the vectors of parameters (i.e., effects) for the different sets of variables, u_{1j} and u_{2j} represent random intercepts to control for unobserved heterogeneity between products. The

random intercepts u_{1j} and u_{2j} are correlated with each other conditional on the independent variables ($X_{ijt}^{purchase}$, X_{ijt}^{return}) but uncorrelated with the within-product effects (e.g., $valence_{jt} - \overline{valence_j}$) because we also included the between-product effects (e.g., $\overline{valence_j}$) (Mundlak 1978):

$$\begin{pmatrix} u_{1j} \\ u_{2j} \end{pmatrix} \Big| X_{ijt}^{purchase}, X_{ijt}^{return} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \gamma\sigma_1\sigma_2 \\ \gamma\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \right] \quad (7)$$

The test for multicollinearity revealed maximum Variance Inflation Factors (VIF) of 2.70 for electronics and 2.95 for furniture. Thus, multicollinearity did not appear to be a concern for our analysis. We estimate the equations separately for electronics and furniture categories.

Results

We first report the effects of OCRs on return decisions to test our hypotheses (Tables 3–6), Tables 3 and 4 provide the results of the main effects model to test H1–H3. Tables 5 and 6 provide the results to test H4 and H5. Finally, in Table 7 we provide simulations to discuss the net effect of OCRs on gross margin based on the model coefficients. As explained in the model section, we test our hypotheses by means of the within-product

Table 3
Effects of OCRs on product purchase and return probability.

	Purchase				Return			
	Electronics		Furniture		Electronics		Furniture	
	Parameter	SE	Parameter	SE	Parameter	SE	Parameter	SE
Within-product effects of OCRs								
$valence_{jt} - \overline{valence_j}$	.059***	(.005)	.069***	(.009)	.056***	(.017)	.070**	(.031)
$volume_{jt} - \overline{volume_j}$	.001***	(.000)	.001*	(.001)	.000	(.000)	−.001	(.002)
$variance_{jt} - \overline{variance_j}$	−.038***	(.005)	−.061***	(.008)	−.005	(.014)	.047*	(.028)
Between-product effects of OCRs[a]								
$\overline{valence_j}$	.057***	(.002)	.040***	(.004)	−086***	(.007)	−.010	(.013)
$\overline{volume_j}$	.003***	(.000)	.000	(.000)	−.005***	(.000)	−.016***	(.001)
$\overline{variance_j}$	−.054***	(.003)	.036***	(.005)	.003	(.009)	−.066***	(.017)
Product characteristics								
$price_{jt} - \overline{price_j}$	−.281***	(.003)	−.198***	(.005)	.097***	(.010)	.111***	(.018)
$YOM_{jt} - \overline{YOM_j}$	.002	(.003)	−.025***	(.005)	.022**	(.010)	.015	(.017)
$\overline{price_j}$	−.091***	(.000)	−.086***	(.001)	−.038***	(.002)	.054***	(.004)
$\overline{YOM_j}$	−.160***	(.001)	−.017***	(.002)	−.065***	(.004)	−.052***	(.007)
$questions_{jt} - \overline{questions_j}$	−.007***	(.001)	.003	(.002)	−.003	(.002)	−.007	(.006)
$frac_answered_{jt} - \overline{frac_answered_j}$	−.010***	(.004)	−.023*	(.012)	.018	(.013)	−.051	(.041)
$new_competitors_{jt}$	.089***	(.004)	.157***	(.011)	.057***	(.012)	.032	(.037)
$\overline{sales_j}$	.005***	(.000)	.056***	(.001)	.002***	(.000)	−.013***	(.003)
$\overline{questions_j}$	−.021***	(.000)	−.017***	(.001)	.017***	(.001)	.026***	(.002)
$\overline{frac_answered_j}$	.034***	(.003)	−.005	(.008)	−.003	(.010)	.006	(.026)
Customer characteristics								
age_{it}	.002***	(.000)	.003***	(.000)	.002***	(.000)	−.001	(.000)
$male_i$	−.085***	(.002)	.063***	(.004)	−.048***	(.005)	−.073***	(.013)
$urbanicity_{it}$	.012***	(.001)	−.004***	(.001)	−.028***	(.002)	−.022***	(.004)
$relationship_length_{it}$	−.063***	(.000)	−.074***	(.001)	−.011***	(.001)	.029***	(.003)
$previous_purchase_{it}$	−.139***	(.002)	−.051***	(.006)	−.067***	(.006)	−.153***	(.021)
$previous_return_{it}$	.050***	(.003)	.056***	(.012)	.446***	(.010)	.501***	(.037)
$order_size_{it}$					.045***	(.003)	.021***	(.004)
Situation-specific variables								
$campaign_t$	.116***	(.003)	.206***	(.007)	.004	(.010)	−.035	(.022)
$holiday_t$	−.005**	(.002)	.026***	(.006)	−.004	(.008)	−.020	(.019)
$category_pageviews_{jt}$	.026***	(.001)	−.046***	(.006)				
Constant	−1.353***	(.010)	−2.068***	(.021)	−1.383***	(.040)	−1.201*	(.119)
γ (random effect correlation)	−.448***	(.003)	−.478***	(.011)				
ρ (error correlation)	.279***	(.015)	−.437***	(.040)				
BIC	3, 659, 177		643, 549					
Number of observations	6, 696, 578		2, 139, 067		552, 619		78, 444	

[a] Variables reflect the average value of the variables over the product's observation period.
* $p < .1$.
** $p < .05$.
*** $p < .01$.

Table 4
Marginal effects of OCRs on purchase and return probability.

	Purchase		Return	
	Electronics	Furniture	Electronics	Furniture
$valence_{jt} - \overline{valence_j}$	9.14%	14.60%	11.16%	10.34%
$volume_{jt} - \overline{volume_j}$	.15%	.20%	.00%	−.14%
$variance_{jt} - \overline{variance_j}$	−5.65%	−11.70%	−0.96%	6.89%

Table 5
Effects of valence × previous purchase interaction on product purchase and return probability.[a]

	Purchase				Return			
	Electronics		Furniture		Electronics		Furniture	
	Parameter	SE	Parameter	SE	Parameter	SE	Parameter	SE
$valence_{jt} - \overline{valence_j}$ if $previous_purchase_{it} = 1$	.063***	(.008)	−.023	(.024)	.036	(.027)	.074	(.082)
$valence_{jt} - \overline{valence_j}$ if $previous_purchase_{it} = 0$	.055***	(.006)	.085**	(.010)	.063***	(.020)	.095***	(.034)
$volume_{jt} - \overline{volume_j}$	.001***	(.000)	.001*	(.001)	.000	(.000)	−.002	(.002)
$variance_{jt} - \overline{variance_j}$	−.035***	(.005)	−.061***	(.008)	−.002	(.015)	.053*	(.029)

[a] To save space, effects of the control variables are not reported.
* $p < .1$.
** $p < .05$.
*** $p < .01$.

Table 6
Effects of valence × price interaction on product purchase and return probability.[a]

	Purchase				Return			
	Electronics		Furniture		Electronics		Furniture	
	Parameter	SE	Parameter	SE	Parameter	SE	Parameter	SE
$valence_{jt} - \overline{valence_j}$	.043***	(.007)	.035**	(.014)	.101***	(.023)	.116**	(.047)
$volume_{jt} - \overline{volume_j}$	.001***	(.000)	.001	(.001)	.000	(.000)	−.001	(.002)
$variance_{jt} - \overline{variance_j}$	−.038***	(.005)	−.059***	(.008)	.008	(.015)	.054*	(.029)
$(valence_{jt} - \overline{valence_j}) \times price_j$	.006***	(.002)	.014***	(.004)	−.025***	(.007)	−.011	(.015)

[a] To save space, effects of the control variables are not reported.
* $p < .1$.
** $p < .05$.
*** $p < .01$.

variation in the OCR characteristics to distinguish the effects of OCRs itself from perceived product quality, popularity, and heterogeneity effects that may be reflected in OCRs. Thus, we consider all reviews available at the moment of purchase and deduct the long term product average. In doing so, we obtain measures that reflect whether the OCR variables at the time of the purchase were above/below the long term average valence, volume and variance.

Table 7
The effect of reviews on net sales and gross margin.

	Scenario	$cumulative\ sales_{jt}$	$cumulative\ returns_{jt}$	$cumulative\ net_sales_{jt}$[b]	$cumulative\ gross_margin_{jt}$
High perceived quality ($\overline{valence_j} = 4.73$)	"High Starting Ratings"	[a]	[a]	[a]	[a]
	"Middle Starting Ratings"	−2.31%	−8.70%	−1.69%	−0.70%
	"Low Starting Ratings"	−5.00%	−13.04%	−4.22%	−2.61%
Medium perceived quality ($\overline{valence_j} = 4.22$)	"High Starting Ratings"	2.61%	6.33%	2.23%	−2.09%
	"Middle Starting Ratings"	[a]	[a]	[a]	[a]
	"Low Starting Ratings"	−3.08%	−8.86%	−2.49%	−0.12%
Low perceived quality ($\overline{valence_j} = 3.78$)	"High Starting Ratings"	6.35%	12.00%	5.02%	−5.53%
	"Middle Starting Ratings"	3.05%	6.67%	2.19%	−1.81%
	"Low Starting Ratings"	[a]	[a]	[a]	[a]

[a] Because the retailer's gross margins and shipping fees are proprietary, we benchmarked the scenarios against the actual review pattern from the empirical data.
[b] $cumulative\ net_sales_{jt} = cumulative\ sales_{jt} - cumulative\ returns_{jt}$.

The effects of OCRs regarding the purchase decision are as expected and in line with previous studies which provides face validity to our results (Babic et al. 2016). Review valence and volume have a positive effect on the purchase decision, whereas review variance reduces the purchase probability (Table 3, left panel). Furthermore, adding the OCR characteristics offers a significant better model fit. The likelihood-ratio tests conclude that the model with OCR variables fits the data significantly better than the model with only the control variables in both categories (electronics: $\chi^2 = 891.14$, $p < .01$; furniture: $\chi^2 = 246.6$, $p < .01$).

Effects of OCRs on Return Decisions

In support of H1, review ratings at the moment of purchase exert a positive, significant effect on return probability (electronics: $\delta = .056$, $p < .01$; furniture: $\delta = .070$, $p = .02$; Table 3, right panel). This means that if the set of reviews available at the moment of purchase is more positive than the long term product average (i.e., $valence_{jt} > \overline{valence_j}$) the return probability becomes higher. The effects of review valence are substantial on both purchase and returns (Table 4). A one-point increase in review valence results in an increased purchase probability of 9.14% (electronics) and 14.60% (furniture). A point increase in valence increases the return probability by 11.16% (electronics) and 10.34% (furniture). This implies that the effect of review ratings available at the moment of purchase goes beyond the moment of purchase and also affects the decision to return the purchased product.

In contrast to H2, we find no significant effect of the number of reviews on the return decision (electronics: $\delta = .000$, $p = .82$; furniture: $\delta = -.001$, $p = .68$). We determine limited support for H3, because the variance in review ratings increase return probability in the furniture category only at a 10% level and it is not significant in the electronics category (electronics: $\delta = -.005$, $p = .74$; furniture: $\delta = .047$, $p = 09$). The results in Table 3, thus, confirm prior research that OCRs influence the purchase decision but extends prior research by demonstrating that review valence in particular increases the product return probability.[6]

In H4, we posit that the effects of valence are weaker for experienced customers (i.e., customers who have recently bought in the specific category at the focal retailer) because they rely less on OCR when forming expectations. To test this hypothesis, we divide customers in two groups: experienced customers who purchased during the last year ($previous_purchase_{it} = 1$) and novice customers who did not make a purchase in the focal category at that retailer during the previous year ($previous_purchase_{it} = 0$). The results indicate that the effect of $valence_{jt} - \overline{valence_j}$ is weaker for experienced customers (Table 5). More specifically, for the return decision the effect of valence is significant only for novice customers (electronics: $\delta = .063$, $p < .01$; furniture: $\delta = .095$, $p < .01$; Table 5, right panel). To formally test if the effect of valence is significantly weaker for experienced customers, we use the joint *Z*-transform test (Whitlock 2005). The joint *Z*-transform test indicates that the effects on purchase and returns are significantly weaker for experienced customers ($Z = 2.19$, $p = 0.01$). Thus, our results show that OCRs are particularly effective for novice customers who have not purchased during the last year at this retailer in the specific product category – in line with H4.

In H5, we posit that the effect of review valence on the return decision is stronger for less expensive products. The interaction effect between $valence_{jt} - \overline{valence_j}$ and $\overline{price_j}$ on the return decision is significantly and negative in the electronics category and is nonsignificant in the furniture category (electronics: $\delta = -.025$, $p < .01$; furniture: $\delta = -.011$, $p < .48$ – see Table 6). We find that the joint *Z*-transform test (Whitlock 2005) across the two categories is significant ($Z = -2.80$, $p < .01$). Thus, we find support for H5 that the effect of valence on the return decision is stronger for cheaper products. For the purchase decision, the effect of valence is stronger for higher priced products (electronics: $\delta = .006$, $p < .01$; furniture: $\delta = .014$, $p < .01$) which is in line with the finding from Floyd et al. (2014).

Control Variables

To control for between-product variation, we use the average values of OCRs per product *j* to account for perceived product quality, popularity, and heterogeneity effects reflected in OCRs (Table 3). The results show that products with a higher average valence over the product life cycle ($\overline{valence_j}$) exhibit a higher purchase probability and a lower return probability in the electronics category which indicates that review ratings also reflect product quality. Products that prompt on average more OCRs ($\overline{volume_j}$) have a significantly higher purchase probability and a lower return probability in the furniture category. Products that prompt on average more variance in OCRs ($\overline{variance_j}$) have a significantly lower purchase probability in the electronics category and a higher purchase and lower return probability in the furniture category. Therefore, one needs to control for these between-product effects to get a more precise estimate for the effect of OCRs on customer purchase and return decision.

Furthermore, our results indicate that various product characteristics affect the return decision. Products with a lower price currently ($price_{jt} - \overline{price_j}$) have a lower return probability and if the product has a higher average price ($\overline{price_j}$) the return probability decreases in the electronics category and increases for furniture. Products longer on the market ($YOM_{jt} - \overline{YOM_j}$) have a higher return probability, albeit significantly only in the electronics category, whereas products that stay on average longer on the market have a lower return probability ($\overline{YOM_j}$). When new competitors are introduced ($new_competitors_{jt}$), the return probability increases significantly for electronics. Finally, for

[6] The effect of review valence on return probability may be moderated by the number of reviews and the variance (Kostyra et al. 2016). Therefore, we estimated a model including the interactions between valence and volume, and valence and variance. These results show that the interaction between valence and volume is positive and the interaction between valence and variance is insignificant in both categories. However, the size of the interaction coefficients are relatively small and hence, the findings converge with the general findings presented in the main text. We also tested whether the effect of valence on the return decision is asymmetric between high ($valence_{jt} > 3.5$) and low ($valence_{jt} \leq 3.5$) valence but the difference in effect on the return decision was insignificant ($Z = -.53$, $p = .60$). We do not detail results here but they are available upon request.

products with more questions ($\overline{questions_j}$), the return probability is higher.

With regard to customer characteristics (Table 3), males and customers who live in rural areas have higher return probabilities. Furthermore, experienced customers ($previous_purchase_{it}$) showed lower purchase and return probabilities, whereas customers who returned during the 12-month period prior to purchase ($previous_return_{it}$) had higher purchase and return probabilities. These findings are consistent with Petersen and Kumar (2009). Detailed results of the control variables can be found in Table 3.

Simulation: Does High Review Valence Mean High Gross Margin?

This study empirically shows that increases in review valence lead to increased product purchase probability as well as product return probability. Hence, depending on the relative sizes of these effects higher review valence may be an advantage or a disadvantage from a retailer's perspective. Therefore, we perform a simulation to enhance the understanding of the net impact of OCRs. This is of high interest to business practitioners, who are eager to know the effect of stimulating customers to post positive reviews on retailer profits (Moe and Trusov 2011). Because several recent studies showed that current reviews affect future reviews due to social dynamics (e.g., Moe and Trusov 2011) and demonstrated feedback loops between purchases and future reviews (Ho-Dac, Carson, and Moore 2013), we also take these interdependencies into account when assessing how increases and decreases in review valence affect retailer profits (see Web Appendix).

To examine the impact of OCRs on retailer's financial performance, we simulate three scenarios with varying initial ratings to illustrate the impact of OCRs on purchases and returns and thereby on retailer performance. In these scenarios, four reviews appear on the day of the product's introduction, with varying star ratings. In Scenario 1, we consider the most positive set of reviews ("High Starting Ratings": valence = 4.75; variance = 0.25). In Scenario 2, the ratings are less positive with an average rating of 4.25 ("Middle Starting Ratings": valence = 4.25; variance = 0.25). Finally, in Scenario 3 we consider the least positive set of reviews ("Low Starting Ratings": valence = 3.75; variance = 0.25). For the purpose of this simulation, we select three actual products from our data to demonstrate the difference between products with low, medium and high $\overline{valence_j}$. For a product with a high perceived quality, we select a monitor which received an average rating of 4.73 over its product life time and an average of 4.75 over the first four reviews which corresponds to the "High Starting Ratings" scenario. For the medium perceived quality product, we select an all-in-one-printer with a $\overline{valence_j} = 4.22$ and an average of 4.25 over the first four reviews which corresponds to the "Medium Starting Ratings" scenario. For the low perceived quality product, we select a notebook with a $\overline{valence_j} = 3.78$ and an average of 3.75 over the first four reviews which corresponds to the "Low Starting Ratings" scenario. We use the observed page view data for these products to simulate the purchase and return decisions in the three different scenarios. We consider the median cumulative sales, cumulative returns and gross margin result of each scenario over 101 simulations (odd number to obtain median) after 100 days.

To compare the scenarios on their profit impact, we calculate the product gross margin (Eq. (9)), which is the difference between the selling price and the costs of goods sold (Farris et al. 2009). In this gross margin, we also include the reverse logistics costs ($reverse_logistics_{jt}$) (Anderson, Karsten, and Simester 2009; Petersen and Kumar 2009), calculated as the sum of the shipping fee, resale discount, and remanufacturing costs (Eq. (8)). The shipping fee, provided by the online retailer, is specific to the product category. The resale discount (15%) and remanufacturing costs (e.g., repackaging, product inspection; 7.5%) both reflect percentages of the retail price and are taken from Guide et al. (2006). In addition, the retailer lost the margin it had earned previously on the purchase (Eq. (9)). We thus compute financial impact as follows:

$$\begin{aligned} &reverse_logistics_{jt} \\ &\quad = shippingfee_j + price_{jt} \\ &\qquad \times (resale_discount_j + remanufacturing_costs_j) \end{aligned} \tag{8}$$

$$\begin{aligned} &gross_margin_{jt} \\ &\quad = margin \times \sum_{t=1}^{T} ((sales_{jt} - returns_{jt}) \times price_{jt}) \\ &\qquad - \sum_{t=1}^{T} (reverse_logistics_{jt} \times returns_{jt}) \end{aligned} \tag{9}$$

where $sales_{jt}$ and $returns_{jt}$ are the numbers of purchases and returns on day t for product j, and $margin$ (%) is the average product margin for the retailer in the electronics category.

We find that the median number of reviews after 100 days is very similar across scenarios, which implies that the differences found between the scenarios are driven by the effect of the star rating of the reviews. Across the three products, the outcome metrics (cumulated across the first 100 days) indicated that the "High Starting Rating" scenario earns the highest cumulative sales but also provokes the most returns (Table 7). The cumulative net sales (i.e., $cumulative\ sales_{jt} - cumulative\ returns_{jt}$) are highest for all products in the "High Starting Rating" scenario. Hence, the effect of having more positive reviews on the purchase probability outweighs the effect on the return probability in terms of net sales. When we account for gross margin, the simulations for the medium ($\overline{valence_j} = 4.22$) and low ($\overline{valence_j} = 3.78$) perceived product quality product show a different pattern than for the high ($\overline{valence_j} = 4.73$) perceived product quality product. For the latter, the "High Starting Rating" scenario has the highest gross margin. For the medium perceived product quality product ($\overline{valence_j} = 4.22$), the "Low Starting Rating" scenario has a -.12% lower margin than the "Medium Starting Rating" scenario (due to lower sales), but the gross margin of the "Middle Starting Rating" scenario is 2.09%

higher than that for the "High Starting Rating" scenario. For the low perceived product quality product ($\overline{valence_j} = 3.78$), this pattern is even clearer. The "Low Starting Rating" scenario (our base scenario) had a 1.81% higher margin than the "Medium Starting Rating" scenario and a 5.53% higher gross margin than that for the "High Starting Rating" scenario.

These results of our simulation demonstrate that if product evaluations are more positive than the product-level average evaluation, sales are relatively high but so are returns, resulting ultimately in lower profits for the retailer. However, these simulations show that highly positive reviews do not always result in lower profits but only when the product reviews at the moment of purchase are more positive than the average review rating over the product life time. In the latter case, the product does not meet these high expectations formed at the moment of purchase, which results in negative expectation disconfirmation and ultimately increases product return probability.

Although the particular results of this simulation are specific to our study context, they provide a clear managerial implication: Marketers should not encourage only very satisfied customers to write reviews (e.g., Kostyra et al. 2016) because the increased purchase probability of overly positive reviews does not offset the negative impact on product return probability. Overall, our results show that every review pattern deviating from the average review rating over the product life time leads to suboptimal performance implications for the retailer.

Discussion

Product returns are currently an important problem for online retailers and have substantial profit impact (Petersen and Kumar 2009, 2015). Consequently, a substantial body of research focused on the impact of product return policies (Janakiraman, Syrdal, and Freling 2016). In addition, prior research highlighted the impact of information provided by retailers on product return probabilities (De, Hu, and Rahman 2013; Shulman, Cunha, and Saint Clair 2015). However, in online retailing, customers base their decision to purchase also on information in OCRs because they provide information about the experiences and perceptions of other costumers and are perceived as trustworthy (Kostyra et al. 2016). Online, customers have incomplete information about the product when purchasing (Wood 2001), and therefore customers will determine whether the product meets their expectations after the product is delivered. Consequently, if the expectations formed at the moment of purchase are too positive due to overly positive ratings (i.e., valence higher than the long term product-average), customers may be more likely to purchase but also the return probability increases due to expectation disconfirmation when the product is finally inspected. Therefore, the aim of this study is to test if OCRs available at the moment of purchase affect the return decision in addition to their effect on the purchase decision.

We test the relation between OCR, purchases and returns by means of a bivariate probit model with endogenous sample selection on a rich multi-year database provided by a major online retailer in two large categories. In our model, we separate between-product heterogeneity reflected in OCRs from within-product variations in OCRs over time (i.e., representing the focal effect of OCRs) using Mundlak's (1978) correction. These detailed analyses produce important findings.

Our main finding is that the OCRs available at the moment of purchase affect both the purchase and return decision. More specifically, if reviews are *overly positive* (i.e., valence is higher than the long term product average), this leads to more purchases but also increases the return probability. This effect is contingent on several factors. First, the effect of review valence on the return decision is weaker for customers experienced in buying at this retailer and in this category. Second, the effect of review valence on the return decision is also weaker for more expensive products. These finding are in line with the expectation disconfirmation mechanism because experienced customers rely less on OCR than novice customers (Chen and Xie 2008). Furthermore, customers spend greater time on making an optimal decision for products with higher prices (Petersen and Kumar 2009) and hence rely less on their expectations formed prior to purchase. Review valence also reflects product quality; products with higher average long-term ratings (i.e., $\overline{valence_j}$) have a higher purchase and lower return probability.

As mentioned in the introduction section, our study is related to the work by Sahoo, Dellarocas, and Srinivasan (2016). They report a negative effect for "valence of reviews", whereas we find a positive effect for $valence_{jt} - \overline{valence_j}$ on the decision to return the product. This can be explained by the difference in operationalization. Their "valence of review" variable (used as a control variable for product quality) is conceptually similar to our $\overline{valence_j}$ variable (i.e., between-product variation), and their findings for "valence of review" are indeed very similar to our findings for $\overline{valence_j}$. Our study shows that the positive effect of review valence on the return decision (i.e., within-product effect) contrasts the product quality effect reflected in review ratings (i.e., between-product effect). This key finding of a positive within-product effect of $valence_{jt} - \overline{valence_j}$ on product returns is missing in the Sahoo, Dellarocas, and Srinivasan (2016) working paper. In addition, our results for volume and variance of OCR are very similar to their results when they include fixed effects for the product (Sahoo, Dellarocas, and Srinivasan 2016, Table 5), which is conceptually very similar to the Mundlak approach we apply. Hence, the results that can be compared between both studies are highly consistent, but our approach provides richer insights by focusing on all three OCR characteristics explicitly and by including both within- and between-product effects.

In line with previous studies, review volume increases the purchase probability, but we find no significant effect of review volume on the return decision. Review variance decreases purchase probabilities; it marginally increases the return probability in one product category. A possible explanation for the non-significant effect of review variance on the return decision is that there are two opposing mechanisms. Higher review variance reduces the expectations due to negativity bias (Chen and Lurie 2013) and therefore reduces the purchase and return probability. On the other hand, review variance increases uncertainty which would lead to lower purchase probability and a higher return probability.

To understand the impact of OCRs on financial performance, researchers must account for their influence on return decisions, and go beyond the effect on just purchases as is the current standard. Incorporating product returns is important because they result in high reverse logistics costs, varying between $6 and $18 per product (The Economist 2013). We further illustrate the net impact of OCRs by means of simulations based on our model coefficients and financial data provided by the retailer. Our simulation demonstrates that if the valence is higher than the long-term average valence for the product (i.e., overly positive reviews), this leads to more purchases but also more returns. Consequently, the substantial reverse logistics costs produce a negative net effect of overly positive reviews on gross margin. That is, the increased purchase probability due to more positive reviews does not offset the negative impact on product return probability, because of the substantial impact of reverse logistics costs.

We think that establishing this link results in a better understanding of the effect of OCRs and drivers of return decisions and thereby contributes to the literature. As current research found mainly positive effects of reviews from a retailer's perspective (Floyd et al. 2014), our study indicates that reviews also may have negative consequences on retailer's profit. Our study challenges the current belief that higher ratings are a universally positive effort as may be derived from the results of several recent meta-analyses (Babic et al. 2016; Floyd et al. 2014). Our study confirms previous studies that review ratings and volume increase sales, but adds to this that overly positive review ratings also increase the number of returns. More specifically, while review valence increases net sales, overly positive review valence impact on financial performance is ultimately negative, because of the high costs associated with product returns.

These findings have important implications for retailers: retailers should provide information that set the right expectations for customers. Currently, many retailers encourage customers to write reviews (Godes and Mayzlin 2009). Given our results, it is important that the reviews reflect all buyers' opinions regarding the product. This is crucial since consumers heavily weight the average rating (i.e., valence) compared to other cues for quality like price and brand when forming quality inferences (De Langhe, Fernbach, and Lichtenstein 2016; Kostyra et al. 2016). Consequently, our findings encourage retailers to get a large review base that adequately reflects the performances of the product. Retailers thus should actively stimulate customers to write a review after purchase, for example by providing them an appropriate incentive (Verlegh et al. 2013) or by reminding them to write a review. However, our results underscore the importance that the reviews are a good representation of the product which needs to be considered when designing an incentive mechanism. In addition, our results indicate that product return decisions are influenced by the information provided about the product at the moment of purchase. Retailers could also provide additional information and tools besides OCR that allow customer to evaluate whether the product fits their needs and set the right expectations (De, Hu, and Rahman 2013; Shulman, Cunha, and Saint Clair 2015).

Finally, we note several routes for extending this work. First, our research is limited to quantitative measures of OCRs. Considering the content, such as the discussion of product features and return decisions, and linguistics style could be a promising area; recent research into the effect of reviews on conversion rates suggests an asymmetric effect of affective content (Ludwig et al. 2013). Second, though we provide some initial generalizations across two large product categories, further research could examine a wider range of product categories, including both search and experience products (Weathers, Sharma, and Wood 2007). Third, in our simulation we show that the effects of OCR vary by the level of $\overline{valence}_j$. Although, this provides important insights on when the effects of OCR are most prominent, future research could use a more direct measure of product quality such as the consumer reports score (De Langhe, Fernbach, and Lichtenstein 2016; Mitra and Golder 2006). Fourth, other sources of information available at the moment of purchase influence product return probability, such as retailer-provided information (e.g., product descriptions, pictures; De, Hu, and Rahman 2013). Comparative analyses could provide valuable insights into whether reviews or retailer-provided information is a more important driver of return rates. However, Kostyra et al. (2016) find that the importance of retailer-provided information decreases if OCRs are available and so we anticipate that the findings will be similar, even with the inclusion of a retailer-provided information variable. Finally, like all other research in that area, our approach does not analyze customers' visual attention on product- and review-specific information in their choice process. We leave these topics for further research and hope our findings stimulate more interest in the important practical topics of OCRs and product returns.

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Appendix A. Supplementary data

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